

PDF Parsing and Source Attribution in RAG

PDF parsing is an important step in many RAG systems because many useful documents are stored as PDF files. Research papers, technical manuals, reports, and policy documents are often distributed in PDF format. A RAG system needs to extract readable text from these files before it can build a knowledge base.

PDF parsing can be difficult. A PDF page may contain headers, footers, page numbers, tables, figures, formulas, and multi-column layouts. If the parser extracts text in the wrong order, the downstream chunks may become noisy. Noisy chunks can hurt retrieval quality because the embedding model receives unclear or fragmented text.

Source attribution is another important requirement. When a chunk comes from a PDF, the metadata should keep the file name and page number. This allows the system to tell the user where the evidence comes from. For example, a source may be shown as `sample_rag.pdf, page 1`. Without source attribution, a RAG answer is harder to verify.

Retrieval Evaluation and Ablation Study

Retrieval evaluation measures whether the retriever can find the correct evidence for a question. A common metric is Recall@K. Recall@K checks whether the expected evidence appears in the top K retrieved chunks. For example, Recall@5 measures how often the correct evidence appears within the first five retrieved results.

A QA benchmark is useful for RAG evaluation. Each item can contain a question, a reference answer, an expected source file, and expected keywords. The expected source file helps check whether the retrieved chunk comes from the correct document. The expected keywords help check whether the retrieved text contains the important evidence.

Ablation study is used to understand which design choices improve a RAG system. Common ablation variables include chunk size, top-k, reranking, and hybrid retrieval. By changing one variable at a time and keeping other variables fixed, we can observe how the retrieval score and answer quality change.

Reranking can improve retrieval quality by reordering initially retrieved chunks. Hybrid retrieval combines vector retrieval and keyword retrieval. Vector retrieval is good for semantic similarity, while keyword retrieval is often better for exact terms, names, numbers, and abbreviations.